

AUTHOR MANUSCRIPT

Probabilistic stacked denoising autoencoder for power system transient stability prediction with wind farms

Please cite the published article

Tong Su, Youbo Liu, Junbo Zhao, Junyong Liu. Probabilistic stacked denoising autoencoder for power system transient stability prediction with wind farms. IEEE Transactions on Power Systems 36 (4), 3786-3789 (2021).

<https://doi.org/10.1109/tpwrs.2020.3043620>

Version and reuse

Publication and download notices have been removed from the manuscript margins; article text and figures are unchanged. This sheet retains the published citation and copyright notice.

© 2020 IEEE. Personal use of this material is permitted. Permission from IEEE must be obtained for all other uses, in any current or future media, including reprinting/republishing this material for advertising or promotional purposes, creating new collective works, for resale or redistribution to servers or lists, or reuse of any copyrighted component of this work in other works.

Publisher sharing policy

Probabilistic Stacked Denoising Autoencoder for Power System Transient Stability Prediction with Wind Farms

Tong Su, Youbo Liu, *Member, IEEE*, Junbo Zhao, *Senior Member, IEEE*, Junyong Liu, *Member, IEEE*

Abstract—To address the uncertainties of renewable energy and loads in transient stability assessment with credible contingencies, this letter proposes a stacked denoising autoencoder (SDAE)-based probabilistic prediction method. The correlations among wind farms have been effectively considered through the variable transformation via the Cholesky decomposition. SDAE allows learning the mapping relationship between operational features and the transient stability margin. The possible operation scenarios are sampled under different confidence levels to generate appropriate inputs for SDAE to assess the probabilistic transient stability distribution. Results on the modified IEEE 39-bus system show that our proposed method can achieve a similar level of accuracy as the benchmark and improved Monte Carlo simulations-based methods while having much higher computational efficiency.

Index Terms—Transient stability, probabilistic prediction, wind uncertainty, stacked denoising autoencoder (SDAE).

I. INTRODUCTION

LARGE scale integration of uncertain and intermittent renewable energy increases the concerns of power system transient stability [1]. Most transient stability assessment approaches are based on the current system operating conditions [2]. To tackle the uncertainties of renewable energy and loads, probabilistic transient stability assessment methods are also developed [3-5]. The online transient stability analysis allows us to take timely actions to maintain system stability. For look-ahead scheduling and planning of power systems, predictions of renewable energy and loads are used to assess if the system can maintain stability. To achieve that goal, the decision tree, support vector machine, and convolutional neural networks are proposed [6-7]. However, the uncertainties of renewable energy and loads are not considered, yielding binary decisions about the system stability with large errors. This letter develops a stacked denoising autoencoder (SDAE)-based probabilistic transient stability prediction method to deal with wind power and load uncertainties considering credible contingencies. The main contribution is that our proposed method can achieve a similar level of accuracy as the original and improved Monte Carlo simulations-based methods while having a much higher computational efficiency. The correlations among renewable energy are also considered. The proposed probabilistic quantification of stability yields a better dynamic situational awareness for system planners and operators and allows them to balance the tradeoff between system stability and economic efficiency during decision makings.

II. PROPOSED PROBABILISTIC STABILITY PREDICTION METHOD

In this section, the modeling of wind power correlation is first presented, followed by the training sample generation and group Lasso-based feature selection. Finally, the SDAE-based probabilistic transient stability prediction is developed.

A. Wind Power Correlation Modeling

The statistical distribution of wind speed follows the Weibull distribution and its probability density function (PDF) is:

$$f(v_w, \lambda, k) = \frac{k}{\lambda} \left(\frac{v_w}{\lambda}\right)^{k-1} e^{-(v_w/\lambda)^k} \quad (1)$$

where $v_w(t)$ denotes the wind speed; k is the shape parameter and λ is the scale parameter of the wind speed. The cumulative distribution function (CDF) for the Weibull distribution is:

$$F(v_w, \lambda, k) = 1 - e^{-(v_w/\lambda)^k} \quad (2)$$

Since two geographically close wind farms correlate with wind

speed, the Cholesky decomposition is used to transform the correlated ones into independent variables when generating sample data. For the correlation matrix $\mathbf{A}$, its Cholesky decomposition is:

$$\mathbf{A} = \mathbf{L}\mathbf{L}^* \quad (3)$$

where $\mathbf{L}$ is a lower triangular matrix with its conjugate transpose $\mathbf{L}^*$. For the wind power with random variable $\mathbf{x}_w$ of correlation matrix $\mathbf{A}$, by using $\mathbf{y}_w = \mathbf{L}\mathbf{x}_w$, the new $\mathbf{y}_w$ will be independent but its covariance matrix is the same as that of $\mathbf{x}_w$. In this letter, the transformed $\mathbf{y}_w$ will be used during the training process.

B. Training Sample Generation and Feature Selection

Training data generation and feature selection are important for deep learning applications. First, we set the fluctuation range of each load and use the Latin hypercube sampling (LHS) to generate load data, i.e., $\mathbf{P}_L = (P_{L1}, P_{L2}, \dots, P_{LN})^T$; the correlated wind speed data $\mathbf{W} = (W_1, W_2, \dots, W_N)^T$ are further used to obtain the corresponding wind power output $\mathbf{P}_w = (P_{w1}, P_{w2}, \dots, P_{wN})^T$. Taking these N initial operation scenarios as the input, the most severe $N-1$ credible contingencies are simulated, e.g., three-phase short-circuit fault transmission line. Assuming the number of lines is L , N by L transient stability simulations are needed to obtain the corresponding generator power angle trajectories. Then, the transient stability index (TSI) is [8]:

$$\text{TSI} = \frac{360 - |\delta^{\max}|}{360 + |\delta^{\max}|} \times 100 \quad (4)$$

where $\delta^{\max}$ is the maximum power angle difference between any two generators. A higher TSI means that the system is more stable while 0 means the boundary. N initial operation scenarios correspond to N by L TSIs, and the minimum TSI is used as the label of a sample for the corresponding initial operation scenario.

Another widely used index, namely the hybrid transient stability margin (HTSM) based on the corrected transient energy function (CTEF) [9] is also introduced and compared with the TSI index in this letter. According to the post-fault trajectories, the generators are divided into two groups: the advanced generator group and the remaining group. The corrected transient kinetic energy (CTKE) is:

$$V_{KE}^{CO} = \frac{1}{2} M_{eq} \omega^2 \quad (5)$$

$$M_{cr} = \sum_{i=1}^{n_{cr}} M_i, \quad M_{sys} = \sum_{i=1}^{n_{sys}} M_i, \quad M_{eq} = \frac{M_{cr} M_{sys}}{M_{cr} + M_{sys}} \quad (6)$$

$$\omega_{cr} = \frac{1}{M_{cr}} \sum_{i=1}^{n_{cr}} M_i \omega_i, \quad \omega_{sys} = \frac{1}{M_{sys}} \sum_{i=1}^{n_{sys}} M_i \omega_i, \quad \omega = \omega_{cr} - \omega_{sys} \quad (7)$$

where V_{KE}^{CO} denotes the CTKE; M is the generator inertia constant; ω is the generator rotor speed; n_{cr} and n_{sys} are the numbers of advanced and remaining generators, respectively. The corrected transient potential energy (CTPE) is:

$$V_{PE}^{CO} = \int_{\alpha^{sep}}^{\alpha} -\hat{P}_{cr-sys} d\alpha \quad (8)$$

where V_{PE}^{CO} denotes the CTPE; $\hat{P}_{cr-sys}$ is the acceleration power between the two generator groups. V_{CTEF} is the sum of CTKE and CTPE, yielding

$$V_{CTEF} = V_{PE}^{CO} + V_{KE}^{CO} \quad (9)$$

If the sign of ω is always positive during the simulation period, the system is transient unstable, which corresponds to $\text{TSI} < 0$. At this

time, CTKE is not completely converted to CTPE, and HTSM is defined as negative CTKE [9]:

$$\text{HTSM} = -V_{KE}^{CO} = -\frac{1}{2} M_{eq} \omega^2 \quad (10)$$

If the sign of ω changes from positive to negative in the simulation period, the system is transient stable, which corresponds to $\text{TSI} > 0$. The HTSM is defined as the increased fault duration from stable to unstable [10]:

$$\text{HTSM} = \Delta T_f \quad (11)$$

For large-scale systems, a lot of data storage is required, and the training burden of deep learning models can be very high. Since some features have less impact on the TSI or HTSM, only the crucial features that have high impacts are selected through feature selection to improve computational efficiency without loss of accuracy. This is achieved via the least absolute shrinkage and selection operator (Lasso)-based regression [11]. It is a regularization method for feature selection and regularization. For the generated N samples, each sample includes m state features and if $\mathbf{x}_i = (x_1, x_2, \dots, x_m)^T$ represents the state feature vector for the i^{th} sample and $\mathbf{y}_i$ represents the TSI or HTSM, the optimization objective of Lasso is:

$$\begin{aligned} \min_{\beta_0, \beta} & \left\{ \frac{1}{N} \|\mathbf{y} - \beta_0 \mathbf{1}_N - \mathbf{X}\beta\|_2^2 \right\} \\ \text{s.t. } & \|\beta\|_1 \leq t \end{aligned} \quad (12)$$

where β_0 and β are the variables to be optimized; t is a pre-specified free parameter; $\mathbf{X}$ is the features matrix; $\mathbf{y}$ is the output matrix corresponding to $\mathbf{X}$; $\|\mathbf{u}\|_p = (\sum_{i=1}^N |u_i|^p)^{1/p}$ is the standard L_p space; $\mathbf{1}_N$ is an N by 1 vector of ones.

After Lasso regression, β is a sparse matrix, and the parameters of those features that have little contribution to the output will be set to 0 and filtered out [11]. As a result, only the most effective and crucial features are retained before training the deep learning model.

C. Stacked Denoising Autoencoder

This letter adopts the SDAE [12] to identify the mapping relationship between the input data and the TSI or HTSM. SDAE is an enhancement of the autoencoder (AE), where AE is a three-layer unsupervised learning model and contains an encoder and a decoder. In the encoding stage, the input data are mapped to the hidden layer via the following mathematical expression:

$$\mathbf{h} = \sigma(\mathbf{W}\mathbf{x} + \mathbf{b}) \quad (13)$$

where $\mathbf{x}$ denotes the input layer vector; $\mathbf{h}$ is the hidden layer vector; σ is the activation function, such as ReLU, Sigmoid, and Tanh; $\mathbf{W}$ represents the weight matrix and $\mathbf{b}$ is the bias vector.

In the decoding stage, the input will be reconstructed via

$$\mathbf{x}' = \sigma'(\mathbf{W}'\mathbf{h} + \mathbf{b}') \quad (14)$$

where $\mathbf{x}'$ is the reconstruction vector of $\mathbf{x}$; σ' , $\mathbf{W}'$ and $\mathbf{b}'$ are the parameters in the decoder.

The optimization objective of the AE is to minimize the loss of reconstruction, namely:

$$L(\mathbf{x}, \mathbf{x}') = \|\mathbf{x} - \mathbf{x}'\|^2 = \|\mathbf{x} - \sigma'(\mathbf{W}'(\sigma(\mathbf{W}\mathbf{x} + \mathbf{b})) + \mathbf{b}')\|^2 \quad (15)$$

Denoising autoencoder (DAE) adds noise to the input data of AE to prevent overfitting and reconstructs the original data by learning the data with noise. The process of adding noise to the input data is:

$$\tilde{\mathbf{x}} \sim q(\tilde{\mathbf{x}} | \mathbf{x}) \quad (16)$$

where $\tilde{\mathbf{x}}$ represents the data with noise; $q(\tilde{\mathbf{x}} | \mathbf{x})$ is an operator to add noise, including Gaussian noise, masking noise, or missing values. The optimization objective of DAE is:

$$L(\mathbf{x}, \mathbf{x}') = \|\mathbf{x} - \sigma'(\mathbf{W}'(\sigma(\mathbf{W}\tilde{\mathbf{x}} + \mathbf{b})) + \mathbf{b}')\|^2 \quad (17)$$

SDAE is the stack of DAEs to reduce computational cost and obtain better abstract ability. The training process of SDAE contains

pre-training and fine-tuning. In the pre-training stage, the first DAE reconstructs input through unsupervised training, and its encoder output is used as the input of the second DAE. By doing this one by one until all DAEs complete the training, the encoders of DAEs are stacked to form SDAE. In the fine-tuning stage, a fully connected layer is stacked at the end of SDAE and trained using labeled data to realize the mapping of input to labels.

D. Probabilistic Transient Stability Prediction

Since different initializations may lead to different conclusions about the system stability assessment, the uncertainties of wind power and loads affect the accuracy of transient stability prediction. Based on current operation scenarios, multiple confidence intervals for the system operating features will be established every 15 minutes with different probabilities [13]. We set four confidence levels with different probabilities, including 90%, 80%, 70%, and 60%. Among them, the 90% probability confidence interval indicates that the power of wind or load has a 90% probability within this interval, and it has the largest interval range among the four intervals. In each confidence level, a large number of possible operation scenarios of the system after 15 minutes are randomly generated and assumed to obey normal distribution. This means that the current active power is the expectation and its standard deviation is the interval range divided by 6 according to the 3-sigma rule. Taking the possible operation scenarios as inputs, the trained SDAE is used to predict the TSI or HTSM. To further subdivide the stability of the system, taking $\text{TSI} = 30$ and 0 or $\text{HTSM} = 0.02$ (seconds) and 0 as the boundary points [8], the system transient stability can be divided into three states: stable, critical stable, and unstable. These values are more conservative as compared to those in [14]. The detailed steps of the proposed probabilistic transient stability prediction method are shown in Fig. 1.

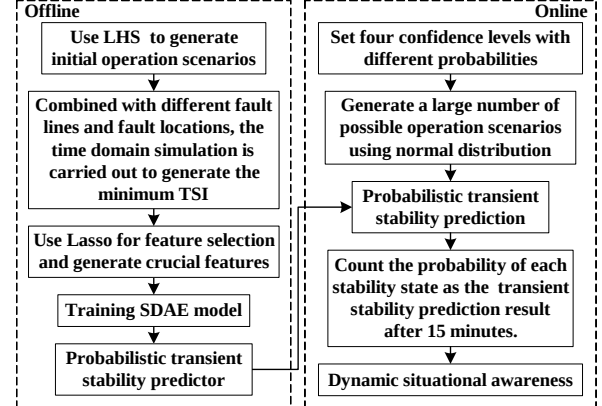


Fig. 1. Proposed probabilistic transient stability prediction method.

III. NUMERICAL RESULTS

Numerical results are carried out on the modified 39-bus system. Two wind farms with a maximum active power output of 500MW at buses 17 and 21 are assumed, where the Doubly Fed Induction Generators (DFIGs) are used to model the wind generator dynamics [5]. They account for 29% of the total loads on those two buses. The active power of wind farms fluctuates from 0 to maximum output. Six different correlation coefficients are set for the wind speeds of the two wind farms, including 0.99, 0.9, 0.6, 0.3, 0 and -0.5. The active power of generators and loads fluctuate in the range of 80% to 120%.

A. Feature Selection

A total number of 6000 initial operation scenarios of the system is simulated, including different fault lines and fault locations for time-domain calculations, yielding 6000 samples for SDAE training. The initial operation scenario $\mathbf{S} = (\mathbf{V}_G, \mathbf{V}_w, \mathbf{P}_G, \mathbf{P}_w, \mathbf{P}_L, \mathbf{Q}_L)^T$ includes the voltage magnitudes of the generators $\mathbf{V}_G$ and wind farms $\mathbf{V}_w$, active power of the generators $\mathbf{P}_G$, wind $\mathbf{P}_w$ and loads $\mathbf{P}_L$, reactive

power of the loads $\mathbf{Q}_L$, leading to a 60-dimension vector. Lasso is used for feature selection and only 24-dimension vectors are selected for transient stability prediction of the power system. The TensorFlow framework is used to build the deep learning model. The SDAE is a 4-layer network stacked by three DAEs and a fully connected layer. In the pre-training stage, the batch size and epoch of DAE are set to 100 and 500, respectively, and 5% of the input data is randomly disturbed as noise. In the fine-tuning stage, the batch size and epoch of SDAE are set to 10 and 500, respectively. The optimizer is Adam, and the activation function is Sigmoid. The offline training time of the proposed probabilistic transient stability prediction is 182s.

When the Lasso is not used, the numbers of neurons in each layer of SDAE are 60-50-20-1. By contrast, when the feature selection by Lasso is leveraged, the numbers of neurons in each layer of SDAE are 24-50-20-1. The accuracy of SDAE using TSI and HTSM are shown in Table I. It can be found that with Lasso, although the prediction accuracy of TSI and HTSM decreases by 0.16% and 1.33%, respectively, the sample dimension decreases by 60% and this significantly reduces the training burden of the SDAE model. When TSI is used, the test set loss with or without Lasso for feature selection is shown in Fig. 2. It can be observed that SDAE can converge rapidly in the early stage of training, and the loss decreases steadily in the later stage without overfitting issues. The deviation of the two loss curves also confirms that there is only 0.16% accuracy loss by using Lasso. In the following, SDAE with Lasso is used for probabilistic transient stability prediction.

TABLE I
ACCURACY OF SDAE WITH DIFFERENT INDICES

Index	Accuracy / %	
	SDAE without Lasso	SDAE with Lasso
TSI	98.66	98.50
HTSM	97.58	96.25

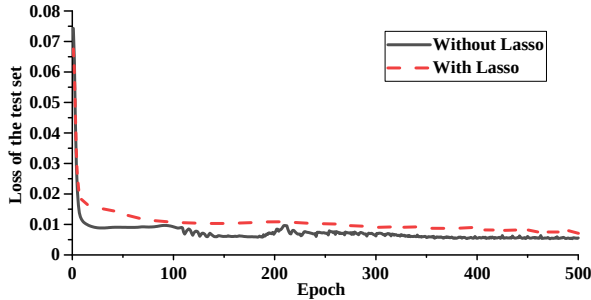


Fig. 2. Loss of test set with or without Lasso regression for feature selection.

B. Probabilistic Prediction Results and Analysis

After the SDAE model structure is determined, based on the current operation scenario, possible operation scenarios of the system considering all credible contingencies after 15 minutes are generated. Under the confidence probabilities of 60%, 70%, 80% and 90%, the active power of the loads and the winds are in the neighborhood with the present active power as the center, and the ranges are 20MW, 30MW, 40MW, and 50MW respectively. The normal distribution is used to generate the random active power distribution, where the expectation is the present active power and the standard deviations are 3.3, 5, 6.7, and 8.3, respectively. In the current operation condition, 1000 possible scenarios are generated corresponding to each confidence probability, yielding a total number of 4000 scenarios. At the 90% confidence level, the transient stability probability distribution results obtained using TSI and HTSM indices via the trained SDAE under five different initial operation scenarios are shown in Table II. It can be found that with the same initial conditions, the transient stable probability distributions for TSI and HTSM are very close. The results also show that the proposed method can adapt to different indices without loss of accuracy. In the following sections,

only TSI is used to demonstrate the performance of the proposed method.

TABLE II
COMPARISON OF TRANSIENT STABILITY PREDICTION RESULTS AFTER 15 MINUTES WITH DIFFERENT INDICES AT 90% CONFIDENCE LEVEL

Initial scenarios	Stable	Probability / %	
		Critical stable	Unstable
TSI = 44.3	100	0	0
HTSM = 0.09	100	0	0
TSI = 15.2	37.9	61.9	0.2
HTSM = 0.01	41.1	58.8	0.1
TSI = -5.1	53.7	45.6	0.7
HTSM = 0.01	70.7	29.3	0
TSI = -25.9	2	63.3	34.7
HTSM = -0.029	1	73.8	25.2
TSI = -64.9	0	0	100
HTSM = -0.57	0	0	100

Table III shows the comparison results for transient stability prediction after 15 minutes under different confident intervals. Note that the TSI for this case is 24.6 that is close to critical stable status. It can be observed that at the 90% confidence level, the probability of transient stable is lower than those of other confidence levels. This is because, at the 90% confidence level, load and wind power have a larger fluctuation range that leads to more unstable scenarios.

TABLE III
IMPACTS OF CURRENT OPERATION CONDITIONS ON SYSTEM TRANSIENT STABILITY AFTER 15 MINUTES AT DIFFERENT CONFIDENCE LEVELS

Confidence level	Transient stability state / %		
	Stable	Critical stable	Unstable
60%	92.4	7.6	0
70%	81.1	18.5	0.4
80%	77.4	19.3	3.3
90%	70.2	23.6	6.2

To further demonstrate that the uncertainties of renewable energy and loads could lead to different transient stability assessment results, the deterministic approach based on load forecasting is compared. It should be noted that accurate load and renewable energy forecasting requires high fidelity information about the weather conditions, including wind speed, light, etc., which are challenging to obtain. As a result, the uncertainty of renewable energy can greatly increase the difficulty of forecasting. Taking the critical stable system in Table III as an example, using the load and wind power forecasting results, the deterministic operation scenario in the next interval is generated, and the TSI is 35.2 from transient stability simulations. This means that the system is stable after 15 minutes. By contrast, the probabilistic prediction shows that the system still has a 6.2% instability risk. This demonstrates the capability of the proposed method is fully recognizing the potential risk and helping planners and operators in developing appropriate preventive controls.

MC has been widely used in the probabilistic transient stability prediction [5], [15] while the Latin hypercube sampling MC (LHSMC) is an improved MC. At the 90% confidence level, 10000 possible operation scenarios are sampled. MC and LHSMC are used to generate transient stability distribution, and their results are used as a benchmark to evaluate the performance of the proposed method, which is shown in Table IV. From Table IV, we can see that with the same initial conditions, the probabilistic prediction results of MC, LHSMC, and the proposed method are close, indicating the good accuracy of the proposed method. However, according to Table V, it can be concluded that the computing times for MC and LHSMC are much higher than the proposed method using different numbers of scenarios. Probabilistic transient stability prediction is an emergent

area and analytical probabilistic methods are not available yet. The reason is that for the analytical methods, accurate distributions of uncertain renewable energy and loads should be known first, which is very challenging in practice. Furthermore, the analytical derivations of the stability index are very complicated. Although being challenging, it could be an important direction to be persuaded in future work.

TABLE IV
COMPARISON OF PROBABILISTIC PREDICTION RESULTS BETWEEN MC, LHSMC AND PROPOSED METHOD (PM)

Initial conditions	Method	Probability / %	
		Stable + Critical Stable	Unstable
TSI = 24.5	PM	52.1	47.9
	MC	55.2	44.8
	LHSMC	55.3	44.7
TSI = 15.8	PM	54.2	45.8
	MC	54.1	45.9
	LHSMC	54.2	45.8
TSI = -14.1	PM	35.3	64.7
	MC	37.6	62.4
	LHSMC	38.4	61.6
TSI = -25.8	PM	39.3	60.7
	MC	39.5	60.5
	LHSMC	39.4	60.6

TABLE V
COMPUTING TIMES FOR MC, LHSMC AND PROPOSED METHOD (PM)

Method	Scenario numbers	Computing times	
		Scenario generations	Transient stability prediction
PM	10000	1.0s	8.6s
MC/LHSMC			>10h
PM	1000	0.1s	3.2s
MC/LHSMC			>1h

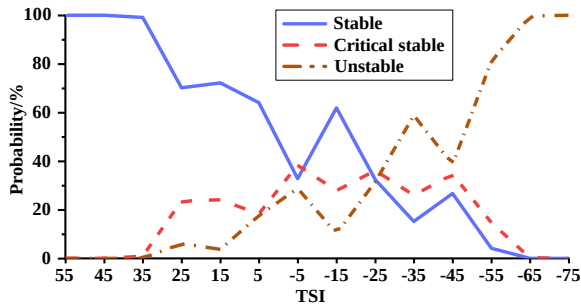


Fig. 3. Comparison of system transient stability after 15 minutes under different current TSIs.

C. Impacts of Current Operation Conditions

To test the impacts of the current TSI on the transient stability prediction results, 14 current scenarios are considered. The TSIs are evenly distributed at 55 ~ -75 at a 90% confidence level. The prediction results after 15 minutes for different current operating conditions are displayed in Fig. 3. It can be observed that with the decrease of the initial TSI, the probability of losing stability is increasing while the critical stability is rising first and then decreasing. This shows that the initial TSI has a great impact on transient stability after 15 minutes. When the system is stable or unstable in the current scenario, it has a large probability of being stable and unstable after 15 minutes. Note that the curves of stable or unstable cases are not monotonously decreasing or increasing. This is because the transient stability prediction is a non-convex problem and two adjacent scenarios in the feature space may have a large difference in transient stability. When the current system operates under critical stable scenarios, all three

transient stability states have a certain probability after 15 minutes. Under this condition, the awareness of transient instability risk via the proposed method is very beneficial.

IV. CONCLUSIONS

This letter proposes a probabilistic SDAE-based transient stability prediction method to address the uncertainties of wind farms and loads considering all possible credible contingencies. Different from the transient stability assessment approaches that deal with the current operation states, the proposed method allows predicting the future system security risks every 15 minutes. This further provides insightful information for the operator to take preventive control for stability enhancement. The key idea is to generate a series of possible operation scenarios by taking into account wind and load uncertainties and feed them into the SDAE-based transient stability predictor. The use of Lasso reduces the training data requirement without loss of prediction accuracy. The test results on the modified IEEE 39-bus system show that the proposed model can obtain a similar level of accuracy as MC and LHSMC while achieving much higher computational efficiency. Future work will include the development of analytical probabilistic transient stability prediction methods.

REFERENCES

- [1] B. Mohandes *et al.*, "A review of power system flexibility with high penetration of renewables," *IEEE Trans. Power Syst.*, vol. 34, no. 4, pp. 3140-3155, Jul. 2019.
- [2] P. Bhui and N. Senroy, "Real-time prediction and control of transient stability using transient energy function," *IEEE Trans. Power Syst.*, vol. 32, no. 2, pp. 923-934, Mar. 2017.
- [3] Z. Yue *et al.*, "Probabilistic transient stability assessment of power system considering wind power uncertainties and correlations," *Int. J. Elect. Power Energy Syst.*, vol. 117, pp. 105649, May 2020.
- [4] P. Ju *et al.*, "Analytical assessment for transient stability under stochastic continuous disturbances," *IEEE Trans. Power Syst.*, vol. 33, no. 2, pp. 2004-2014, Mar. 2017.
- [5] P. Papadopoulos *et al.*, "Probabilistic framework for transient stability assessment of power systems with high penetration of renewable generation," *IEEE Trans. Power Syst.*, vol. 32, no. 4, pp. 3078-3088, Jul. 2017.
- [6] R. Yan *et al.*, "Fast transient stability batch assessment using cascaded convolutional neural networks," *IEEE Trans. Power Syst.*, vol. 34, no. 4, pp. 2802-2813, Jul. 2019.
- [7] Y. Zhou *et al.*, "A novel data-driven approach for transient stability prediction of power systems considering the operational variability," *Int. J. Elect. Power Energy Syst.*, vol. 107, pp. 379-394, May 2019.
- [8] Y. Xu, Z. Y. Dong, L. Guan, "Preventive dynamic security control of power systems based on pattern discovery technique," *IEEE Trans. Power Syst.*, vol. 27, no. 3, pp. 1236-1244, Aug. 2012.
- [9] D. Fang *et al.*, "Transient stability limit conditions analysis using a corrected transient energy function approach," *IEEE Trans. Power Syst.*, vol. 15, no. 2, pp. 804-810, May 2000.
- [10] C. K. Tang, C. E. Graham and M. El-Kady *et al.*, "Transient stability index from conventional time domain simulation," *IEEE Trans. Power Syst.*, vol. 9, no. 3, pp. 1524-1530, Aug. 1994.
- [11] W. Yang *et al.*, "MRM-Lasso: A sparse multiview feature selection method via low-rank analysis," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 26, no. 11, pp. 2801-2815, Nov. 2015.
- [12] P. Vincent *et al.*, "Stacked denoising autoencoders: Learning useful representations in a deep network with a local denoising criterion," *J. Mach. Learn. Res.*, vol. 11, no. 12, pp. 3371-3408, 2010.
- [13] J. Yan *et al.*, "Novel cost model for balancing wind power forecasting uncertainty," *IEEE Trans. Energy Convers.*, vol. 32, no. 1, pp. 318-329, Mar. 2017.
- [14] C. Liu *et al.*, "A systematic approach for dynamic security assessment and the corresponding preventive control scheme based on decision trees," *IEEE Trans. Power Syst.*, vol. 29, no. 2, pp. 717-730, Mar. 2014.
- [15] T. Guo and J. V. Milanović, "Probabilistic framework for assessing the accuracy of data mining tool for online prediction of transient stability," *IEEE Trans. Power Syst.*, vol. 29, no. 1, pp. 377-385, Jan. 2014.